

This section translates the role definitions into technical JSON configurations that control agent behavior within the Kiro CLI system. Participants create properly structured agent files using the official schema, configuring tools access (`read`, `write`, `shell`), implementing security boundaries through `toolsSettings` with path restrictions and command filtering, and linking context files as resources. Each agent receives carefully crafted permissions that enforce role separation—Product Managers can only write requirements, QA Testers have `pytest` capabilities, and Python Developers get full development environment access—ensuring realistic team dynamics through technical constraints.

Product Manager Agent

1. Create `.kiro/agents/product-manager.json`:

```
1  {
2    "name": "product-manager",
3    "description": "Product Manager for requirements definition",
4    "tools": ["read", "write"],
5    "allowedTools": ["read", "write"],
6    "toolsSettings": {
7      "write": {
8        "deniedPaths": [
9          "**/contexts/**",
10         "**/tests/**",
11         "**/src/**",
12         "**/shared/bugs.md",
13         "**/shared/status.md",
14         "**/shared/test-report.md",
15         "**/shared/uat-feedback.md"
16       ],
17       "allowedPaths": [
18         "**/shared/requirements.md"
19       ]
20     }
21   },
22   "resources": [
23     "file://contexts/product-manager.md"
24   ]
25 }
```

QA Tester Agent

1. Create `.kiro/agents/qa-tester.json`:

```
1  {
2    "name": "qa-tester",
3    "description": "QA tester for web application functionality",
4    "tools": ["read", "write", "shell"],
5    "allowedTools": ["read", "write"],
6    "toolsSettings": {
7      "write": {
8        "deniedPaths": [
9          "**/contexts/**",
10         "**/src/**",
11         "**/shared/requirements.md",
12         "**/shared/status.md",
13         "**/shared/uat-feedback.md"
14       ],
15       "allowedPaths": [
16         "**/tests/**",
17         "**/shared/test-report.md",
18         "**/shared/bugs.md"
19       ]
20     }
21   },
22   "resources": [
23     "file://contexts/qa-tester.md"
24   ]
25 }
```

Python Developer Agent

1. Create `.kiro/agents/python-developer.json`:

```
1  {
2    "name": "python-developer",
3    "description": "Python developer for web application development",
4    "tools": ["read", "write", "shell"],
5    "allowedTools": ["read", "write"],
6    "toolsSettings": {
7      "write": {
8        "deniedPaths": [
9          "**/contexts/**",
10         "**/tests/**",
11         "**/shared/bugs.md",
12         "**/shared/requirements.md",
13         "**/shared/test-report.md",
14         "**/shared/uat-feedback.md"
15       ],
16       "allowedPaths": [
17         "**/shared/status.md",
18         "**/src/**"
19       ]
20     }
21   },
22   "resources": [
23     "file://contexts/python-developer.md"
24   ]
25 }
```

✔ Checkpoint: Validate Agent Configurations

1. Check all agent files exist

```
1  # Expected: 3 .json files (product-manager.json, qa-tester.json, python-developer.json)
2  ls -la .kiro/agents/
```

2. Validate JSON syntax

```
1  # Expected: no syntax errors
2  kiro-cli agent validate --path .kiro/agents/product-manager.json
3  kiro-cli agent validate --path .kiro/agents/qa-tester.json
4  kiro-cli agent validate --path .kiro/agents/python-developer.json
```

3. Test agent loading (should load without errors)

```
1  # Expected: Help text showing chat options
2  kiro-cli chat --agent product-manager --help
```

4. List available agents

```
1  # Expected: Shows your 3 custom agents
2  kiro-cli agent list
```

5. Validate custom agent restrictions

- Product Manager

```
1  kiro-cli chat --agent product-manager
```

Prompt:

```
1  Create a Flask app in src/app.py with the following code...
```

Expected: Product Manager should refuse and say they only write requirements, not code. This demonstrates proper role separation.

- QA Tester

```
1  kiro-cli chat --agent qa-tester
```

Prompt:

```
1  Update the requirements in shared/requirements.md to add a new feature
```

Expected: Permission denied - QA Tester can only write test reports, bugs, and test files

🛡 Security Note

If the QA agent tries to create alternative files like `shared/requirements.txt` or `shared/requirements.doc`, this demonstrates a security bypass. The current `allowedPaths` model only restricts specific file paths, not file patterns. In a real-world scenario, you would need additional security measures to prevent agents from creating alternative versions of restricted files.

- Python Developer

```
1  kiro-cli chat --agent python-developer
```

Prompt:

```
1  The requirements in shared/requirements.md are unclear, please update them to be more specific
```

Expected: Permission denied - Developer cannot modify requirements, only implement them